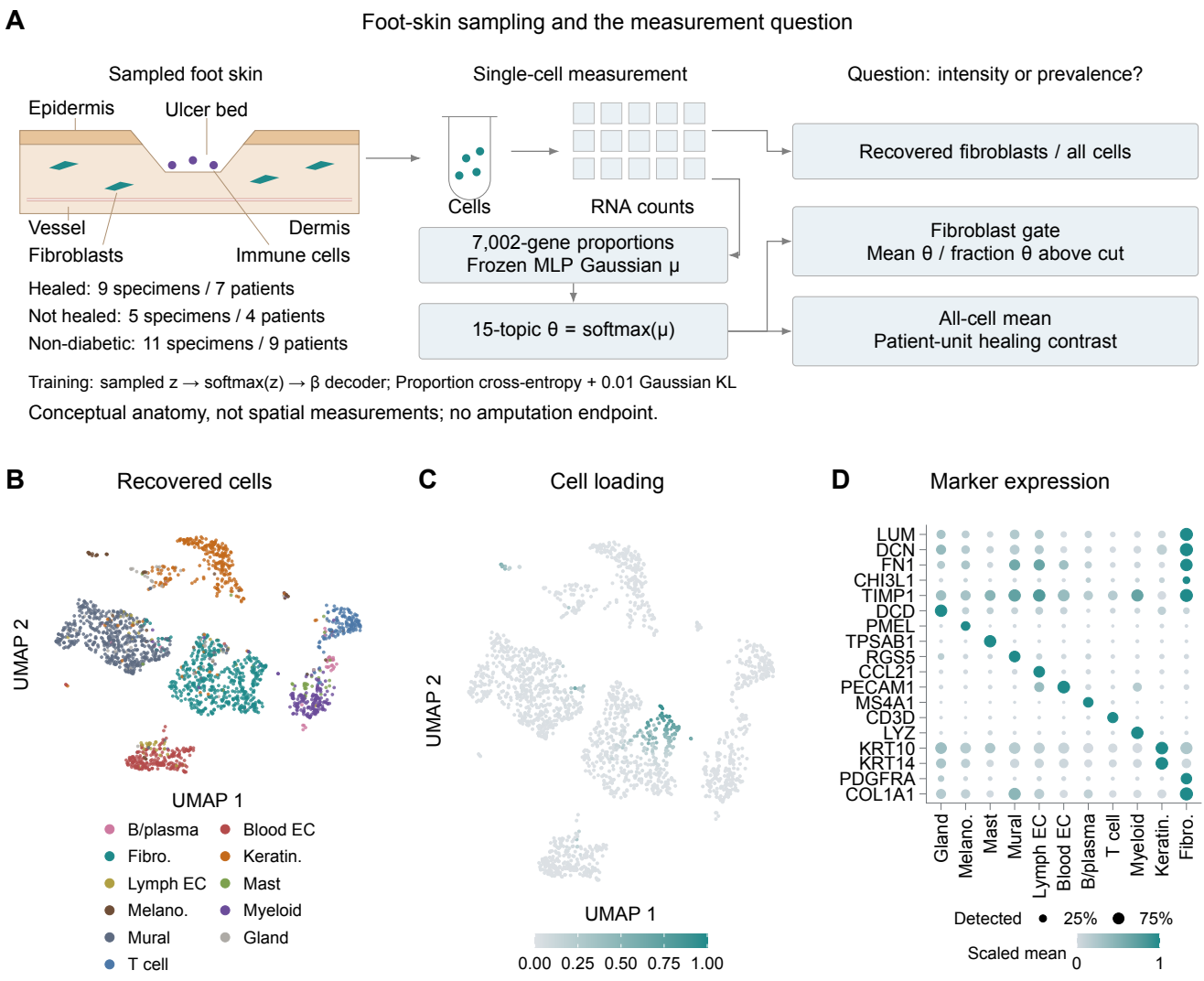
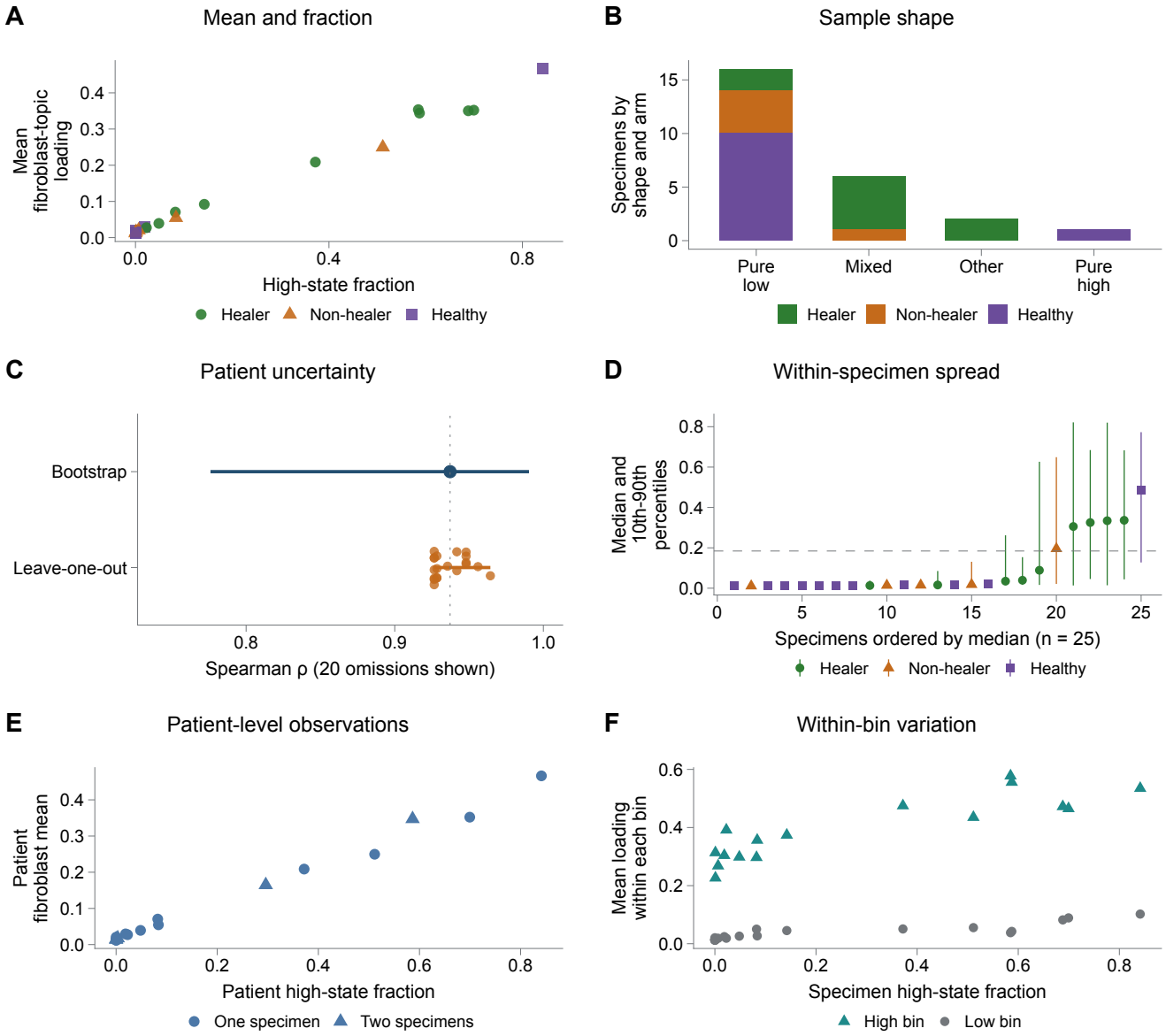
Figure 1



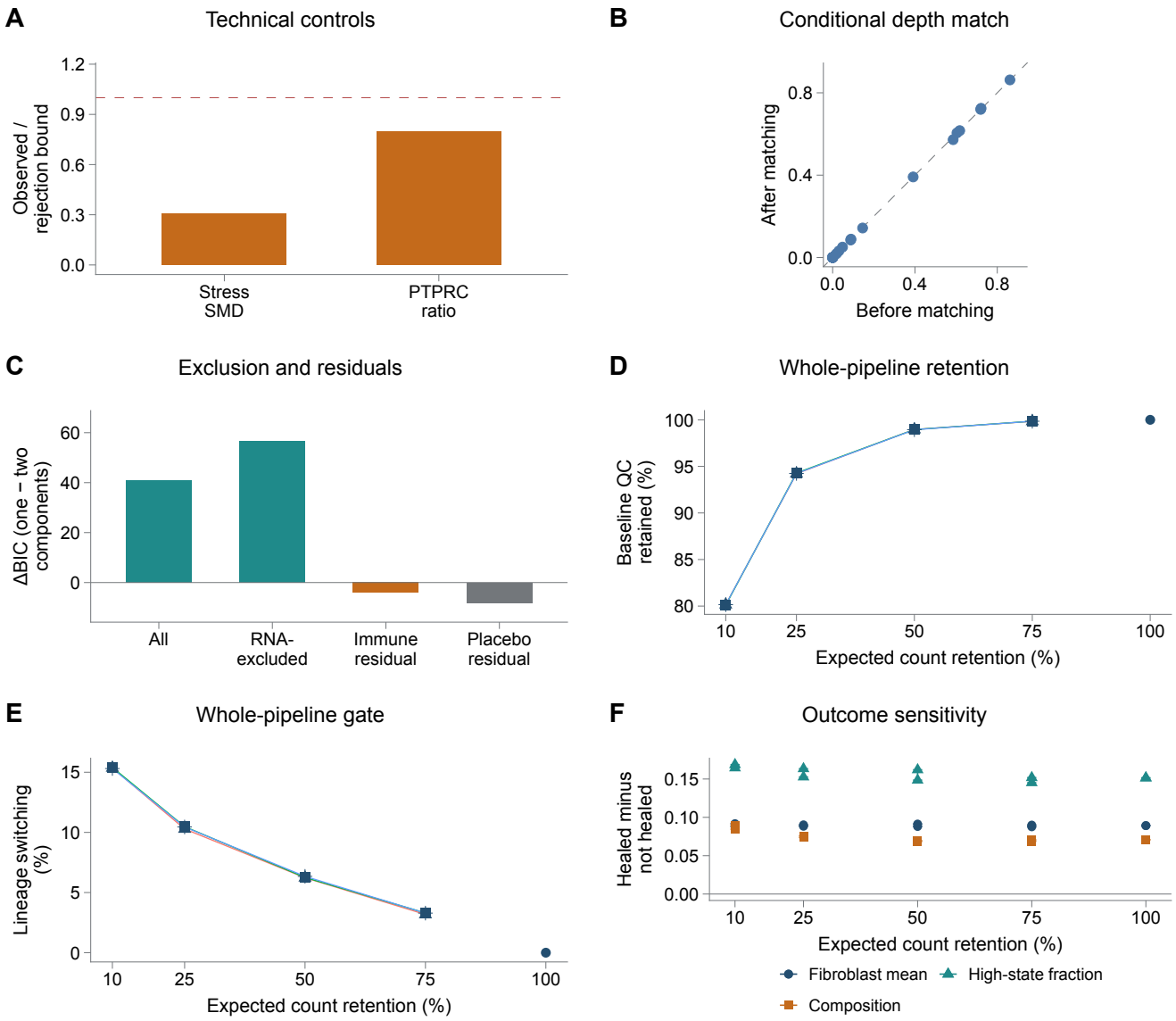
What is measured in diabetic-foot skin? (A) Conceptual foot skin and destructive sampling, not measured histology. The 25 specimens represent 7 healed, 4 non-healed and 9 non-diabetic people. Composition uses recovered-cell counts; frozen inference maps 7,002-gene proportions through Gaussian mean μ to 15-topic softmax θ. Fibroblast mean/fraction and the all-cell patient mean are separate summaries. Neither readout feeds the other. Training: sampled z, normalized decoder β, cross-entropy plus 0.01 Gaussian KL. No amputation endpoint is measured. (B) Uniform display of 2,000 of 76,461 cells by marker-derived lineage; Fibro., fibroblast; EC, endothelial; mural, pericyte/smooth muscle. (C) The same cells by loading. (D) Expression: size is detection fraction, color gene-wise scaled mean log-normalized expression. Annotation markers are not independent label validation. (E) All 19,410 fibroblast loadings with the original cut; inset expands density 0–2.5, preserving the spike. (F) Composition in 25 specimens; colors match B. Non-diabetic people lack healing outcomes; inference uses full data and patient units.

Figure 2



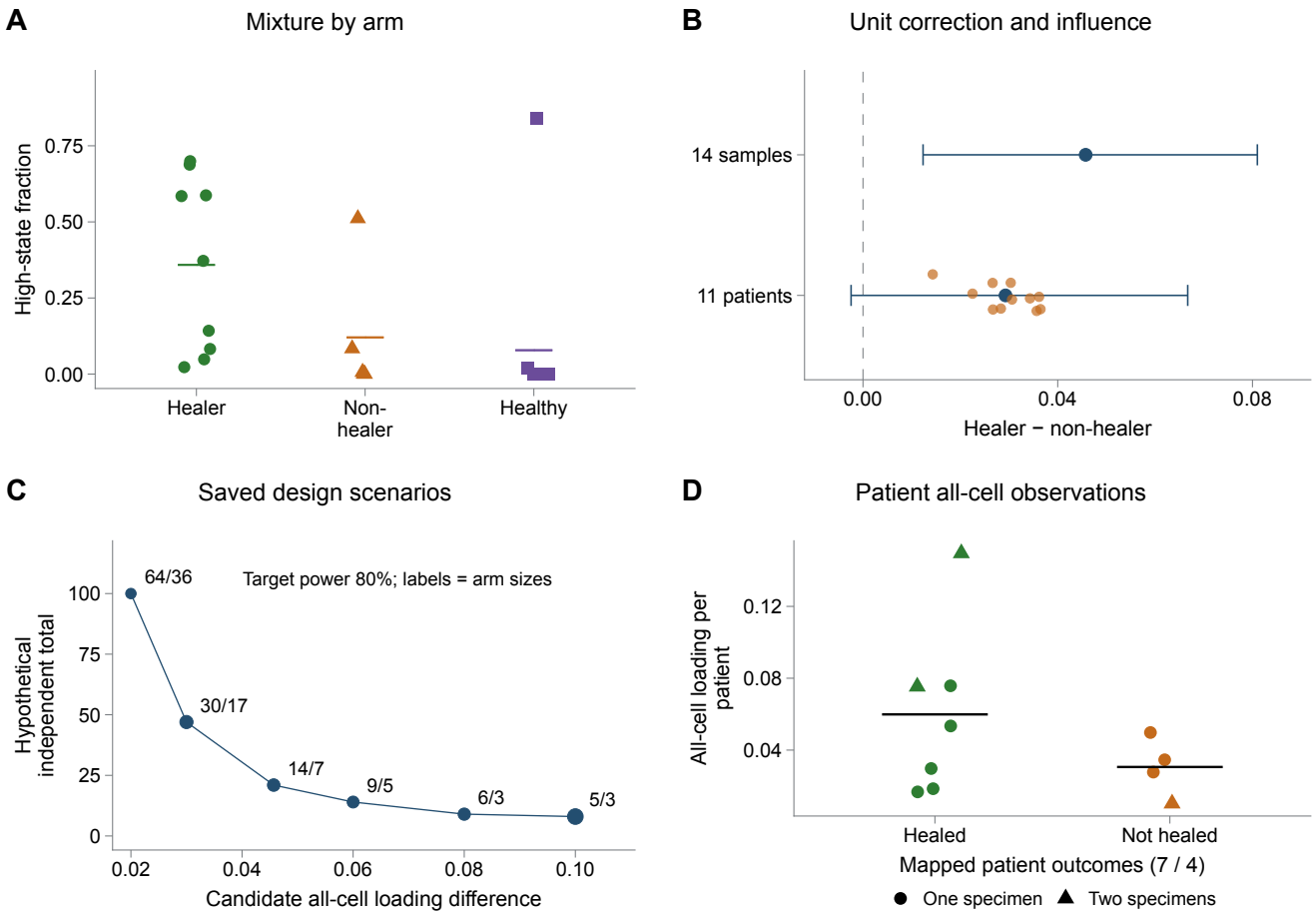
Mean, prevalence and within-bin variation. (A) The 25 specimens comprise 9 healing, 5 non-healing and 11 non-diabetic samples; the cut is 0.1842958 and Spearman ρ=0.933. (B) Stacks retain clinical-arm composition in quantile-defined shapes: pure low has the 90th percentile at most 0.15, pure high the 10th percentile at least 0.10, and mixed the 10th percentile at most 0.05 and 90th percentile at least 0.40. (C) After fibroblast-count pooling to 20 patients, ρ=0.937; blue is the saved patient-bootstrap 95% interval, while orange shows all 20 omission estimates and their range. Vertical jitter separates identical correlations. (D) Ordered specimen medians and 10th–90th percentiles are loading distributions, not confidence intervals. (E) All 20 patient means and fractions; symbols identify one or two contributing specimens, not outcome groups. (F) Low- and high-bin means against specimen high fraction; undefined empty-bin means are not plotted. These shared-cell summaries do not identify discrete biological states, constant component intensities or a healing effect.

Figure 3



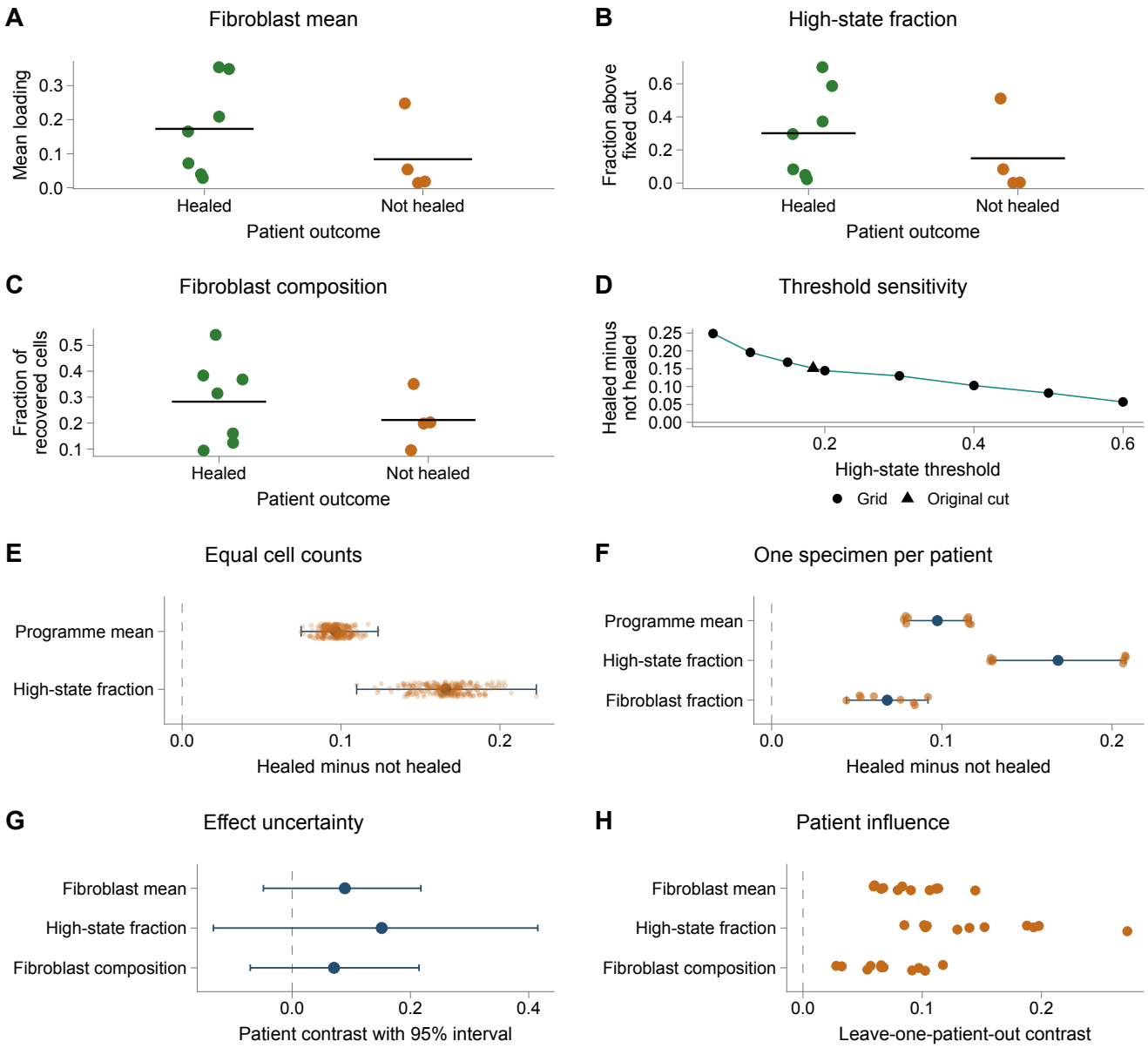
Conditional controls and whole-pipeline sensitivity. (A) Stress SMD and PTPRC rate ratio divided by their rejection bounds; dashed line: one. (B) Specimen fractions before and after matching 14,561 eligible fibroblasts to 986 panel counts; identity is dashed. The 99.5% agreement conditions on original lineage labels. (C) One-minus-two-component BIC for 24 matched specimens before and after immune-transcript exclusion, and separately for immune- and placebo-covariate residuals in 25 specimens. Both residual comparisons favor one component; their samples and perturbations are not interchangeable. (D) Retention of baseline QC survivors after count thinning and rerunning QC. (E) Lineage switches among common survivors. D/E retain all three computational seeds and the unthinned baseline. (F) Seed minima and maxima of patient contrasts for three readouts; these ranges are sensitivities, not confidence intervals. The same 7 healed and 4 non-healed patients, frozen gate, encoder and cut are used. Conditional stability does not establish whole-pipeline invariance, identity truth or clinical portability.

Figure 4



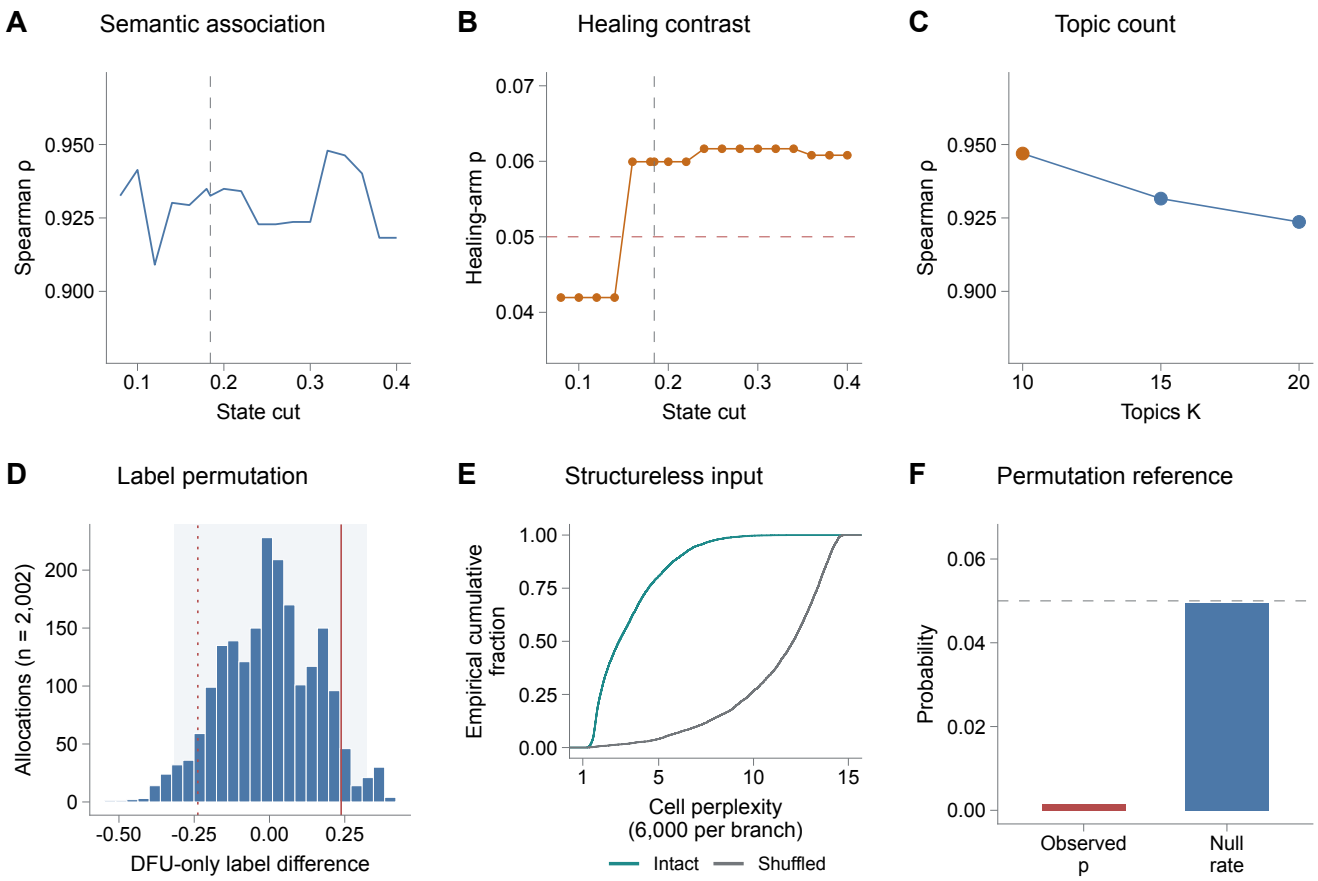
State abundance, analysis unit and saved design scenarios. (A) High-state fractions for 9 healing, 5 non-healing and 11 healthy specimens, with arm means. The healthy maximum 0.841 challenges exclusivity to healing ulcers. (B) All-cell mean-loading effects and bootstrap 95% intervals for 14 specimens versus 11 patients; orange points retain all 11 patient omissions, not additional outcomes. (C) Six previously calculated all-cell loading-difference scenarios targeting 80% power; labels show hypothetical healed/non-healed arm sizes. The specimen-derived spreads and independent-observation assumption are not patient-population variance estimates. These scenarios differ from the historical 24.9% power calculation for a 9-versus-5 high-state-fraction design and are not clinical minimum important differences. (D) All-cell scores of the 7 healed and 4 non-healed patients, with arm means and symbols for one or two contributing specimens. A and D have different cell denominators; none of these displays supplies independent prognostic validation.

Figure 5



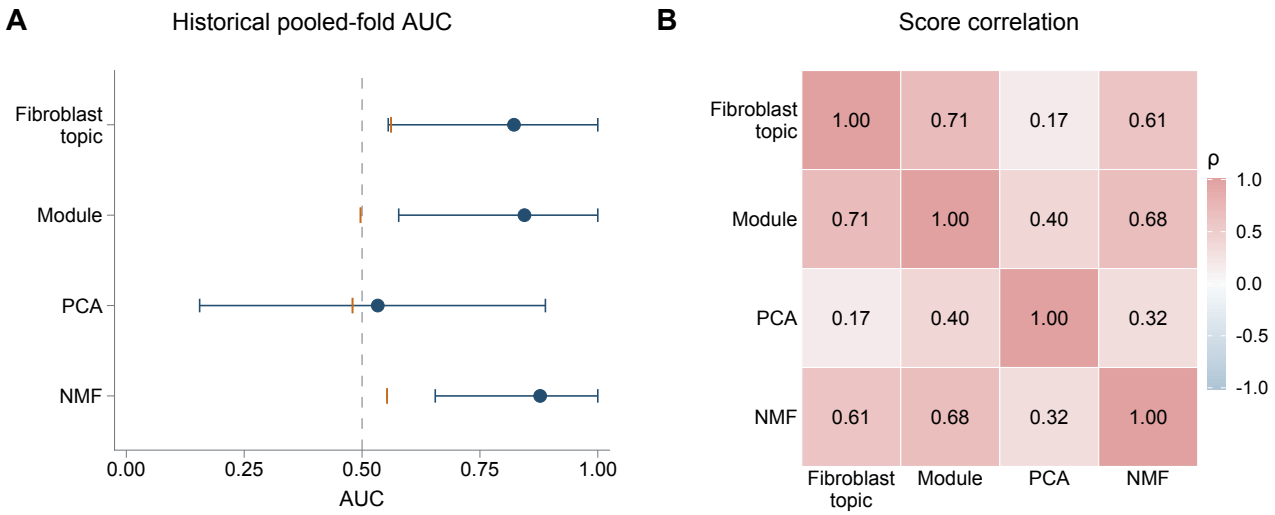
Patient-unit readouts, uncertainty and influence. A–C show 7 healed and 4 non-healed DFU patients, with group means; repeated specimens are pooled using the relevant cell counts. (A) Mean fibroblast loading: difference 0.0892, exact p=0.3121. (B) Fraction above the frozen cut: difference 0.1514, p=0.3697. (C) Fibroblast fraction: difference 0.0708, p=0.4697. Tests use all 330 label allocations; BH q=0.4697 for all three. (D) High-state contrasts across eight fixed cuts and the original midpoint. (E) All 250 saved equal-cell draw contrasts, their means and full ranges, using 100 fibroblasts per specimen. (F) All eight one-specimen-per-patient choices, with means and full ranges. Vertical jitter separates saved values without changing their contrasts. (G) Cell-count-weighted effects with 10,000-draw patient-bootstrap 95% intervals; all span zero. (H) All 11 patient omissions per readout. E, F and H describe the same patients, not confidence intervals or independent replications; positive omission estimates and intervals spanning zero can coexist without validating prognosis or equivalence.

Figure 6



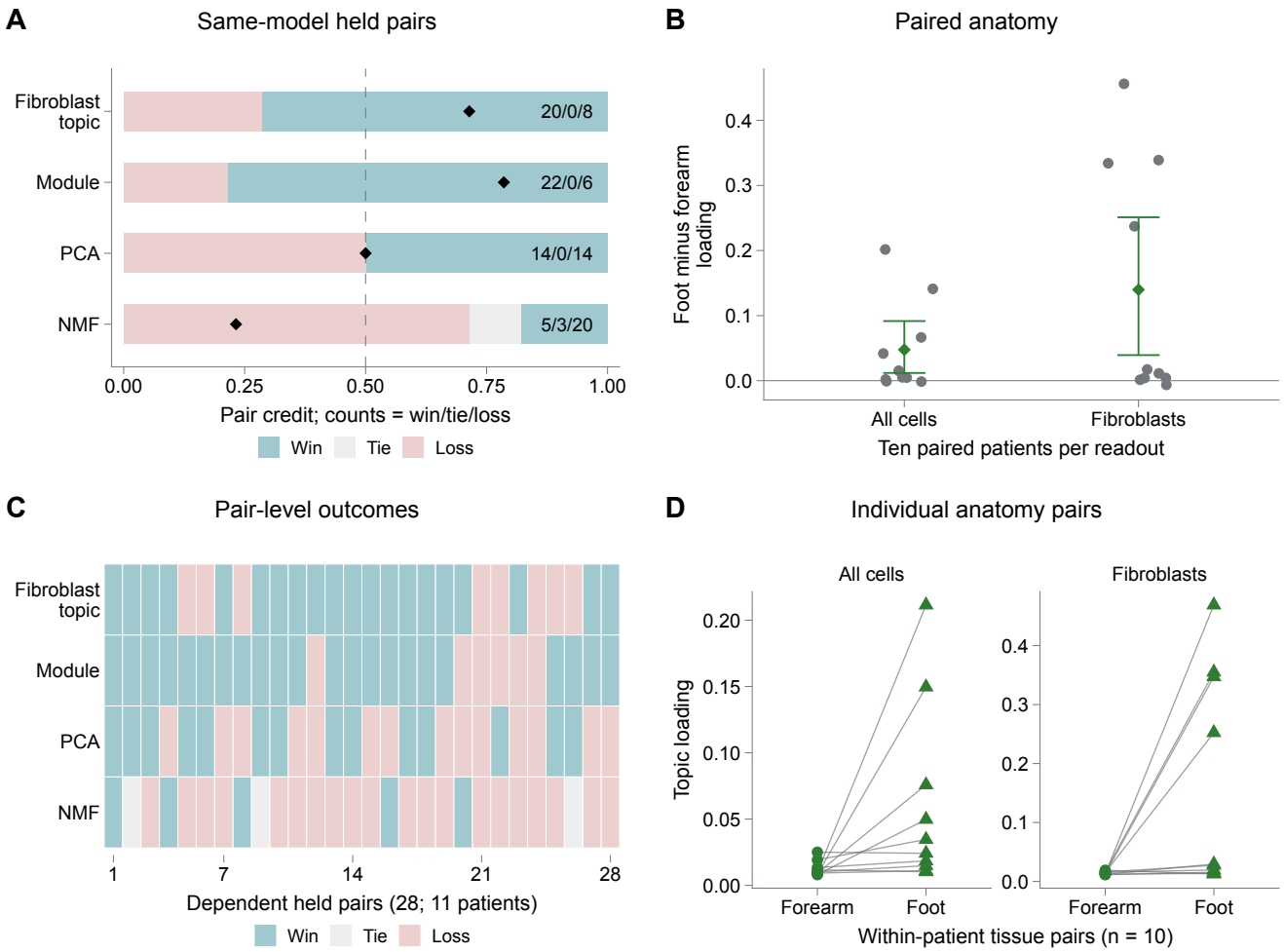
Threshold, capacity and negative-control analyses. (A) Mean–fraction correlation at 18 cuts across 25 specimens; dashed line: pooled cut. (B) Healing-arm Mann–Whitney probabilities for 9 versus 5 specimens; 4/18 are below 0.05. (C) Correlations at K=10, 15 and 20; orange marks K=10, where the healthy-skin maximum fails. (D) The complete saved distribution of 2,002 DFU-only label allocations, with central 95% shading, positive observed difference and its negative counterpart for the two-sided comparison (p=0.1199). (E) Empirical cumulative perplexity distributions of the 6,000 intact and 6,000 gene-shuffled cells. Their mean perplexities are 3.44 and 11.23 of 15; shuffling also changes row-library variation, so this is not a complete technical null. (F) The separate anatomical specimen-level permutation reference: observed p=0.0015 and 5.0% of permuted probabilities below 0.05. Cells and label allocations are computational units, not extra patients; repeated specimens limit inference, and no panel establishes pipeline-wide error control.

Figure 7



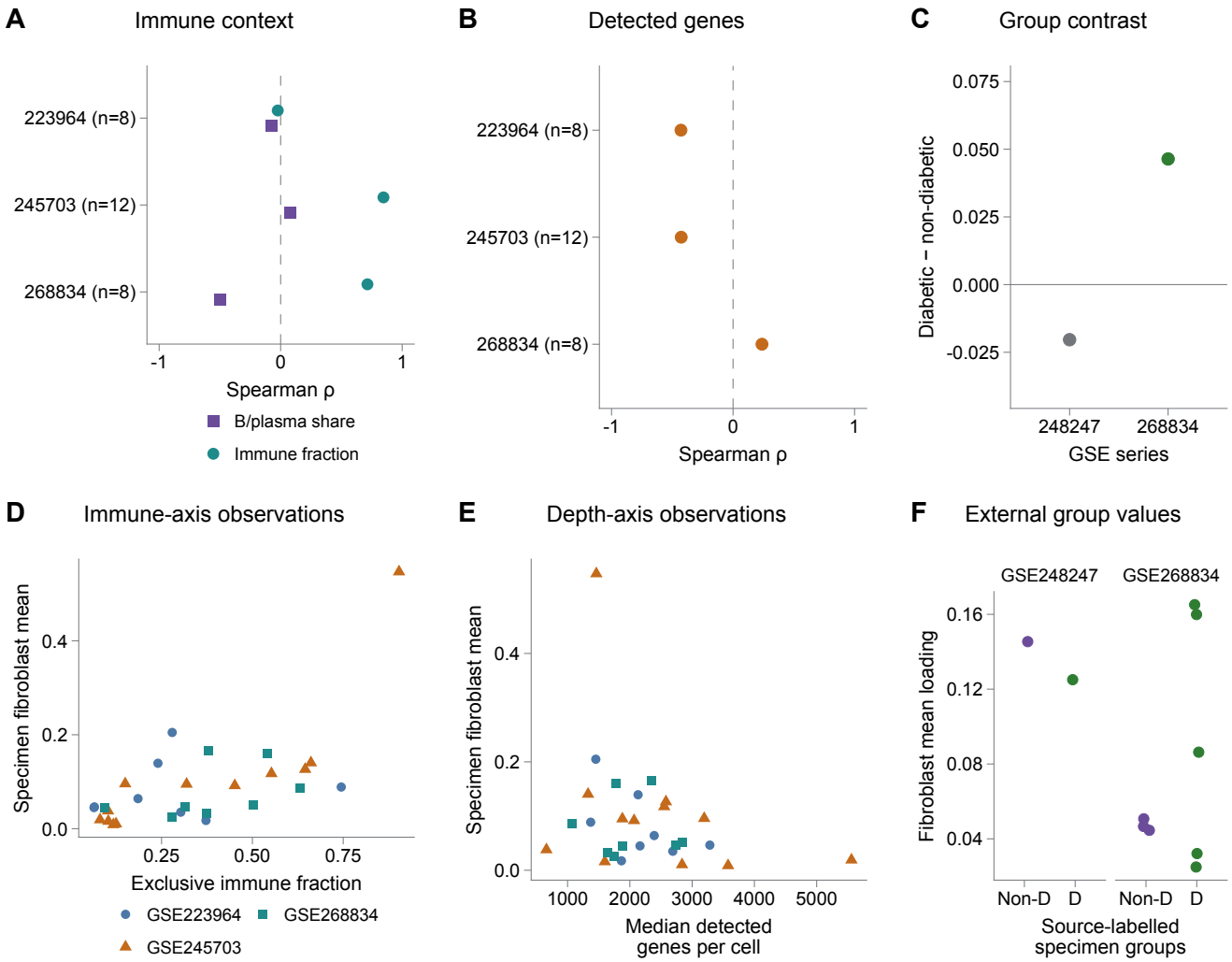
Historical scoring diagnostics in 14 DFU specimens (9 healing, 5 non-healing). (A) Leave-one-sample-out ROC areas (AUCs) with percentile 95% intervals from 4,000 arm-stratified bootstrap draws of fixed out-of-fold scores, using common resampling indices. Orange ticks: permutation-null means after rerunning outcome-dependent fitting and orientation; dashed line: 0.5, not the fitted null. Fibroblast topic is the frozen all-cell mean; Module is the specified fibro-inflammatory score; PCA and NMF denote principal component and nonnegative matrix readouts. Different fitted-fold coordinates and repeated patients prevent these pooled AUCs or intervals from supporting a representation ranking. (B) Spearman correlations of the same 14 specimen readouts; colours and numbers encode the same statistic. These are procedure diagnostics, not independent patient validation.

Figure 8



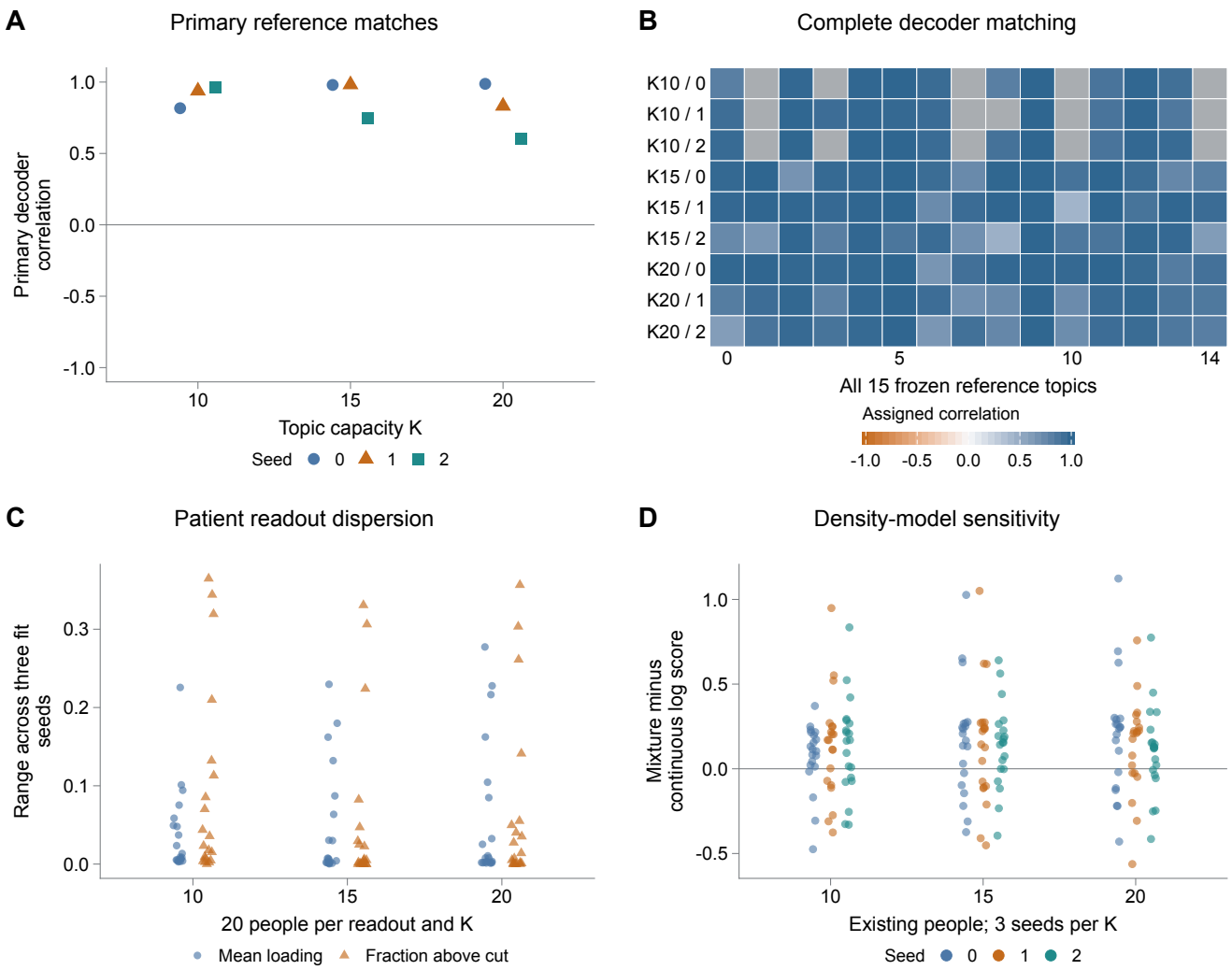
Patient-pair discrimination and anatomical sensitivity. (A) Win/tie/loss fractions and counts for each scoring method, with a black diamond at its mean credit across 28 held pairs. Every fold fits on the other nine patients; ties earn half credit. The pairs share 7 healed and 4 non-healed patients and have no binomial confidence interval. (B) All 10 paired foot-minus-forearm differences for both all-cell and fibroblast-restricted loading, overlaid with saved means and paired-patient bootstrap 95% intervals. For the all-cell topic, sign-flip p=0.0107 and BH q=0.0402. (C) All dependent held-pair wins, ties and losses in saved order. (D) The same 10 patients' forearm and foot levels, separately for the two cell denominators with distinct vertical scales. B and D are different summaries of the same pairs, not 20 independent people. Neither this anatomical sensitivity nor the shared-patient scoring comparison supplies independent clinical validation.

Figure 9



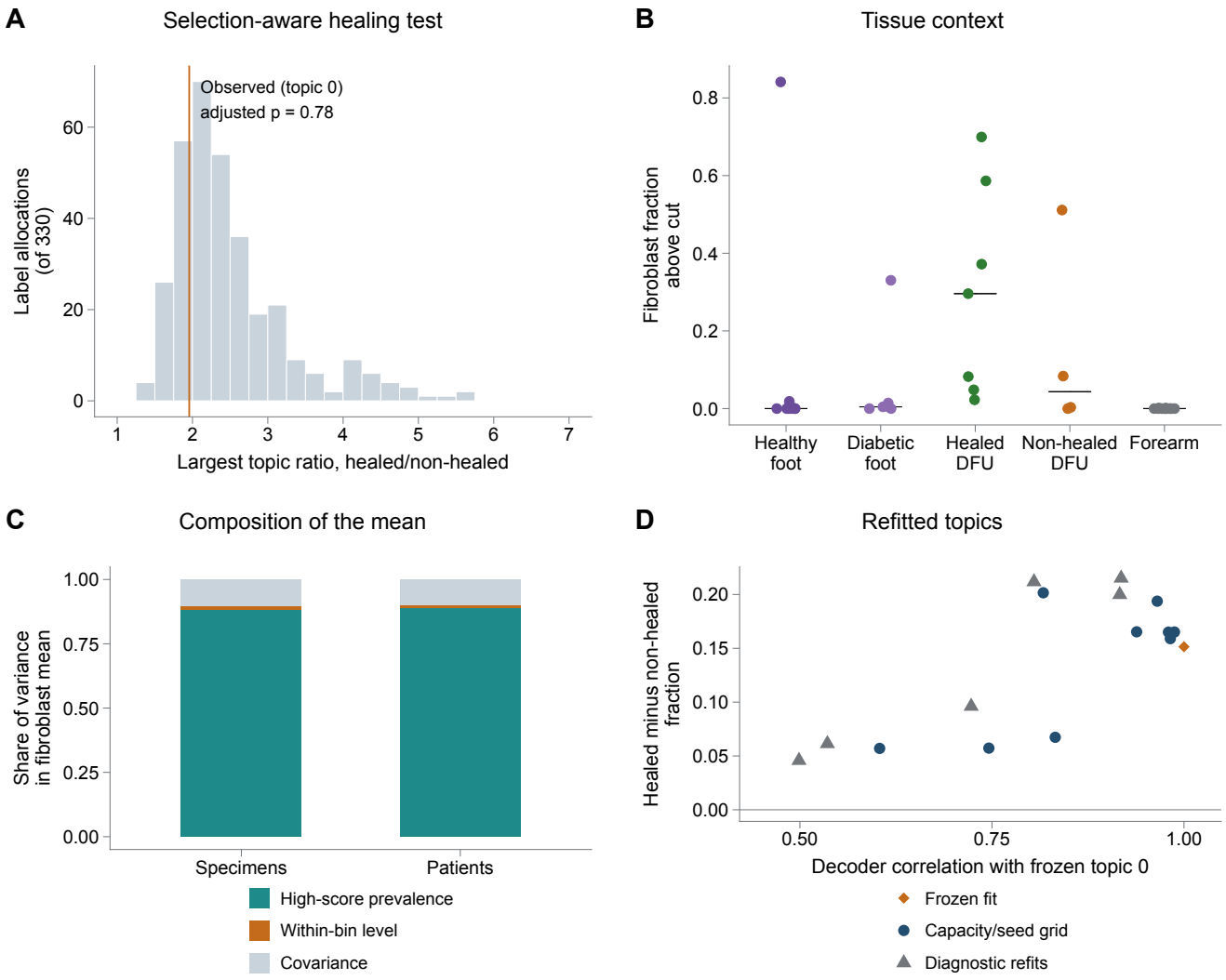
External tissue-context projections, not healing validation. (A) Saved correlations with exclusive immune fraction and B/plasma share. (B) Saved correlations with median detected genes per cell. (C) Diabetic-minus-non-diabetic fibroblast contrasts: +0.0464 in GSE268834 and −0.0204 in GSE248247. (D) All specimen fibroblast means against exclusive immune fractions in the three cohorts (8, 12 and 8 specimens). (E) The same observations against median detected genes per cell. (F) Individual source-labelled diabetic and non-diabetic specimen values in the two group comparisons; GSE248247 has only one specimen per group. Correlations and raw points summarize the same data, not separate replications. Gene coverage, tissue sampling and cell yields differ, authoritative healing outcomes are absent, and no biological confidence intervals are inferred.

Figure 10



Prespecified capacity and training-seed sensitivities in the original discovery cohort. (A) All nine 40,000-step fits retain their one-to-one assigned match to frozen reference topic 0, without favorable-seed selection. Correlation compares square-root decoder gene probabilities; a missing primary match is unavailable rather than rescued with a rowwise best match. (B) Every reference-topic assignment is shown; gray cells are unmatched or undefined, not low-valued assigned correlations. Capacity K=10 necessarily leaves at least five of the 15 reference topics unmatched. (C) Each point is one existing person's range across three seeds at a fixed K, separately for mean loading and fraction above the fit-specific fibroblast cut. The plotted population is fibroblast-only; complete all-cell counterparts and unavailable counts accompany the tables. Equal-specimen means within patient define this sensitivity experiment, not the primary cell-pooled outcome estimand. (D) Points are person-averaged held-cell log-score differences for a two-component mixture versus a continuous density, on fit-specific transformed topic scores. Positive values favor the mixture under that predictive density comparison. Cell splits and threshold exclusions do not make the full-discovery trained representation independent. Seeds, capacities and points do not add biological patients or validate a discrete state, healing biomarker or amputation endpoint.

Figure 11



Selection, tissue context and composition of the fibroblast readout. (A) Each of the 330 allocations of 7 healed among the 11 DFU patients gives the largest healed/non-healed all-cell mean ratio over the 15 topics; the orange line is the observed ratio of topic 0, which was named after its arm-wise means were inspected, and the adjusted p is the fraction of allocations at least as extreme. (B) Patient-level fraction of fibroblasts above the frozen cut in intact healthy and diabetic foot skin, healed and non-healed ulcers, and forearm skin from all groups; bars are medians. (C) Shares of between-unit variance in the fibroblast mean from the high-score prevalence term, the within-bin level term and their covariance in the exact mean decomposition. (D) Healed minus non-healed fibroblast fraction for the frozen fit and 15 refits, each scored with its own matched topic and cut, against the decoder correlation of that topic with frozen topic 0; no refit gives an exact permutation p below 0.05.